Exp No: 04

STUDY OF LOCAL AREA NETWORK USING A SWITCH AND ETHERNET CABLES

AIM:

To set up and configure a Local Area Network (LAN) using a switch and ethernet cables, and verify communication between connected computer.

INTRODUCTION:

A Local Area Network (LAN) is a network that connect computers, and other devices within a limited geographical area such as laboratory, school and home. It enables users to share resources like files, printers and internet connections.

LAN provides reliable, secure and cost-effective communication between devices. A network switch acts as the central connecting device, allowing all connected computers to communicate with each other at high speed.

PROCEDURE:

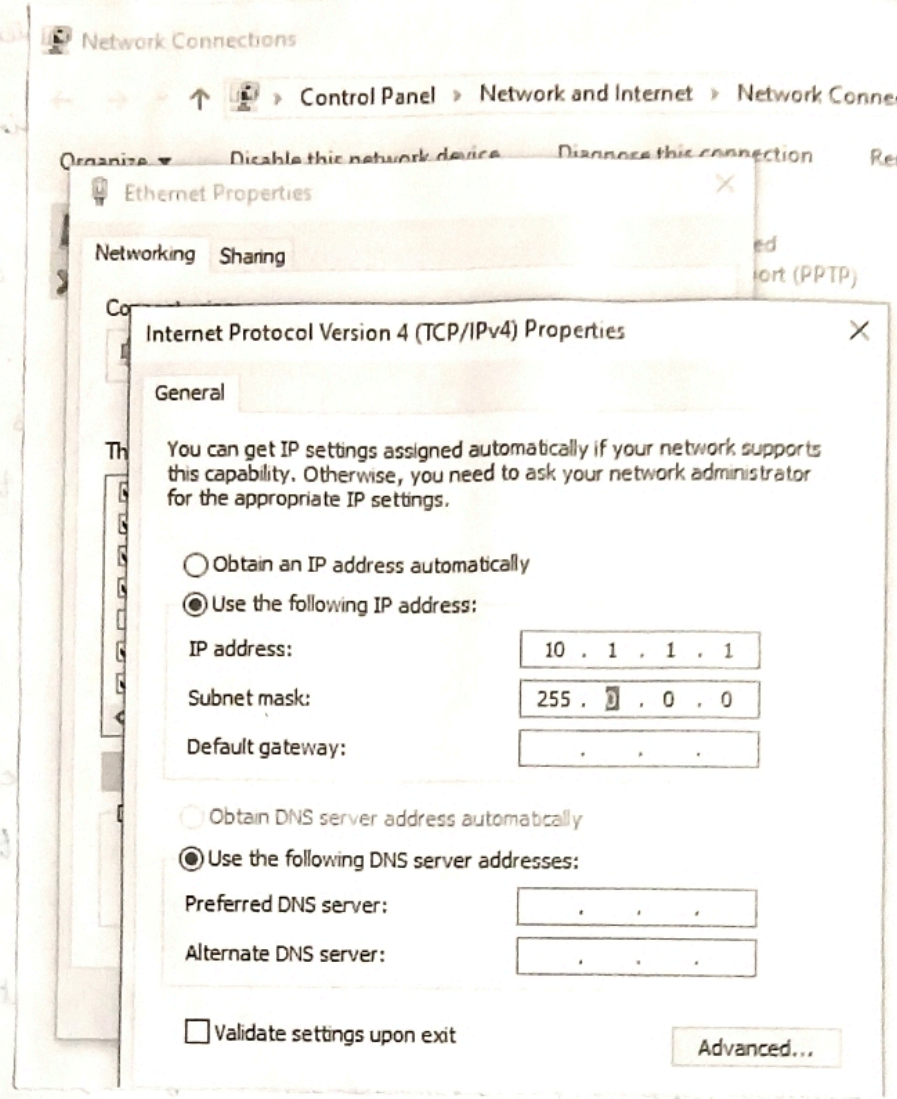
* Plan the network by arranging 4 computers, one network switch, and ethernet cable.
* Connect each computer to the switch using ethernet cables
* Open control Panel → Network → Internet → Network connection.
* Right-Click Ethernet → Properties → TCP/IP v4 → Properties.
* Select use the following IP address and assign.

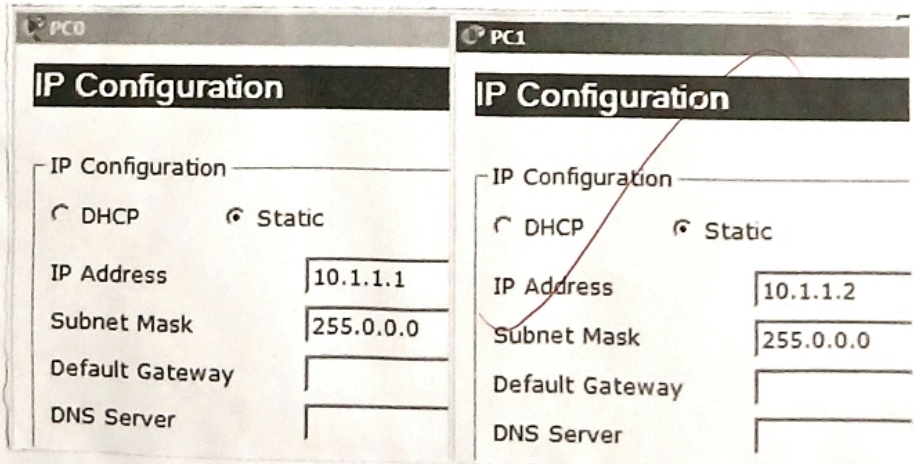
- PC1 : 10:1.1.1, subnet Mask : 225.0.0.0
- PC2 : 10:1.1.2, subnet Mask : 225.0.0.0
- PC3 : 10:1.1.3, subnet Mask : 225.0.0.0
- PC4 : 10:1.1.4, subnet Mask : 225.0.0.0

* Configure the switch by connecting it to a computer, accessing its web interface and assigning the switch IP as 10.1.1.5 with 255.0.0.0

RESULT:

Thus, A Local Area Network (LAN) was successfully set up and configured using a network switch and ethernet cables.


Network Connections
Control Panel
Network and Internet
Network Connec
Organize
Disable this network device
Diagnose this connection
Ethernet Properties
Networking
Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 1
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...


PC0
IP Configuration
IP Configuration
DHCP
Static
IP Address
10.1.1.1
Subnet Mask
255.0.0.0
Default Gateway
DNS Server
PC1
IP Configuration
IP Configuration
DHCP
Static
IP Address
10.1.1.2
Subnet Mask
255.0.0.0
Default Gateway
DNS Server

